

MycoFlow — Biologically Inspired, Reflexive QoS for OpenWrt

GRADUATION PROJECT · 2025-2026

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MOTIVATION

Problem

- **Bufferbloat:** A single TCP flow causes 693 ms queue delay on a 20 Mbit/s WAN connection — making gaming and VoIP unusable.
- **Static QoS:** Even modern AQMs like CAKE rely on DSCP tags; ISPs strip these tags, causing all traffic to fall into the same queue.
- **Fixed Bandwidth:** WAN speed limit is configured once during setup; it cannot react to real-time congestion signals.

APPROACH

Solution

- 1

Biological Inspiration: Adaptive nutrient distribution strategy of fungal mycelium networks → reflexive network control
- 2

6-Class Device Persona: VOIP, GAMING, VIDEO, STREAMING, BULK, TORRENT — On-chip <1 μs decision without DPI or ML

 Gaming  Streaming  VoIP  Bulk  Torrent  Unknown
- 3

12-Class Flow Taxonomy: 3-signal weighted voting combining passive DNS (0.6), target port hints (0.3), and behavioral fingerprinting (0.1)
- 4

Zero Flash Writes: All state is kept in RAM — no damage to NAND flash lifespan (live tested on Xiaomi AX3000T)

COMPARISON

System	Embedded	Automatic	No ML	No DPI	Adaptive
SQM/CAKE	✓	✗	✓	✓	✗
qosify	✓	✗	✓	✗	✗
nDPI	✗	✓	✗	✗	✗
DeepFlow	✗	✓	✗	✗	✗
MycoFlow	✓	✓	✓	✓	✓

DSCP → CAKE TIN MAPPING

 VOIP → EF → Voice tin	 GAMING → CS4 → Voice tin
 VIDEO → CS3 → Video tin	 STREAMING → CS2 → Video tin
 BULK → CS1 → Bulk tin	 TORRENT → CS1 → Bulk tin

SYSTEM SPECIFICATIONS

TARGET HARDWARE
Xiaomi AX3000T (MT7981B, 256MB)

CPU USAGE
<2% (A53)

BINARY SIZE
<200 KB

TEST COVERAGE
13/13 CTest

DEPLOYMENT
OpenWrt IPK

SYSTEM ARCHITECTURE

1 Hz Reflexive Control Loop

🏠 Home Network (LAN) ⇄ **Xiaomi AX3000T + MycoFlow** ⇄ 🌐 Internet (WAN)

① SENSE (Environment Sensing)

ICMP multi-ping (RTT/jitter) · /proc/net/nf_conntrack · Netlink qdisc stats · Passive DNS listening · BPF TC hook passive TCP RTT

ICMP PROBE · CONNTRACK · NETLINK · PASSIVE DNS · BPF RTT

② INFER (Nutrient Density Estimation)

EWMA smoothing · 6-class device persona (majority voting) · 3-signal flow classifier (DNS:0.6 + port:0.3 + behavior:0.1) · 2-tick stability gate

6-CLASS PERSONA · 12-CLASS SERVICE · EWMA A=0.30 · HYSTERESIS

③ ACT (Network Adaptation)

CAKE AQM target update (tc qdisc change) · iptables mangle DSCP rules · libnetfilter_conntrack CONNMARK push · nftables POSTROUTING DSCP restore · IFB ingress shaping

CAKE DIFFSERV4 · DSCP MARK · CONNMARK · IFB

④ STABILIZE (Balancing Reflex)

Feedback ring (8 slots) · Step size adaptation · Sliding EWMA baseline (β=0.01) · RTT auto-demote/promote ladder

FEEDBACK RING · SLIDING BASELINE · RTT DEMOTE

🔄 Feedback to next loop — all state in RAM, zero flash write guarantee

EXPERIMENTAL RESULTS

Core Metrics



154×

Gaming tin latency advantage — Voice 225 μs vs Bulk 34.7 ms



+40.6

DNS signal accuracy increase (pp) — 42.88% → 83.54% offline validation



66.97%

In-vivo tin-matched accuracy (DNS active, 80-catalog, N=1,650 flows)



<200 KB

aarch64 binary size — zero flash writes, musl libc static linking

PHASE 1 — DUAL PERSONA VALIDATION (20 MBIT/S WAN)

FIFO (no QoS)		693 ms
CAKE (standalone)		0.63 ms
CAKE + MycoFlow		0.71 ms (+0.08ms overhead)

PHASE 2 — SIX PERSONA MIXED LOAD (5 CLIENTS)

Waveform Score: A+

CAKE (5 clients)		1.43 ms
MycoFlow (5 clients)		1.10 ms

IN-VIVO DNS VALIDATION — TIN-MATCHED ACCURACY (N=1,650 Flows · seed=42)

No DNS (port+behav.)		49.26%
DNS active (80-cat)		66.97%
DNS active (585-cat)		70.41%

Future Work: Flow-timing analysis for QUIC/ECH is planned.

Conclusion: MycoFlow proves the full viability of biologically inspired reflexive network management (mycelium network concept) on standard home hardware (Xiaomi AX3000T) with zero flash writes and sub-microsecond decision making.

